Special Publication No. 20-12

Documentation of Chinook, Chum, and Coho Salmon Presence in the Upper Goodpaster River Gold Mining District

by

Andrew D. Gryska

December 2020

Alaska Department of Fish and Game Divisions of Sport Fish and Commercial Fisheries

Symbols and Abbreviations

The following symbols and abbreviations, and others approved for the Système International d'Unités (SI), are used without definition in the following reports by the Divisions of Sport Fish and of Commercial Fisheries: Fishery Manuscripts, Fishery Data Series Reports, Fishery Management Reports, and Special Publications. All others, including deviations from definitions listed below, are noted in the text at first mention, as well as in the titles or footnotes of tables, and in figure or figure captions.

Weights and measures (metric)

centimeter	cm
deciliter	dL
gram	g
hectare	ha
kilogram	kg
kilometer	km
liter	L
meter	m
milliliter	mL
millimeter	mm

Weights and measures (English)

cubic feet per second	ft^3/s
foot	ft
gallon	gal
inch	in
mile	mi
nautical mile	nmi
ounce	oz
pound	lb
quart	qt
yard	yd

Time and temperature

day	d
degrees Celsius	°C
degrees Fahrenheit	°F
degrees kelvin	K
hour	h
minute	min
second	s

Physics and chemistry

all atomic symbols	
alternating current	AC
ampere	A
calorie	cal
direct current	DC
hertz	Hz
horsepower	hp
hydrogen ion activity (negative log of)	pH
parts per million	ppm
parts per thousand	ppt, ‰
volts	V
watts	W

General

Alaska Administrative Code	AAC
all commonly accepted abbreviations	e.g., Mr., Mrs., AM, PM, etc.
all commonly accepted professional titles	e.g., Dr., Ph.D., R.N., etc.
at	@
compass directions:	
east	E
north	N
south	S
west	W
copyright	©
corporate suffixes:	
Company	Co.
Corporation	Corp.
Incorporated	Inc.
Limited	Ltd.
District of Columbia	D.C.
et alii (and others)	et al.
et cetera (and so forth)	etc.
exempli gratia (for example)	e.g.
Federal Information Code	FIC
id est (that is)	i.e.
latitude or longitude	lat or long
monetary symbols (U.S.)	\$, ¢
months (tables and figures): first three letters	Jan,...,Dec
registered trademark	®
trademark	™
United States (adjective)	U.S.
United States of America (noun)	USA
U.S.C.	United States Code
U.S. state	use two-letter abbreviations (e.g., AK, WA)

Mathematics, statistics

all standard mathematical signs, symbols and abbreviations	
alternate hypothesis	H_A
base of natural logarithm	*e*
catch per unit effort	CPUE
coefficient of variation	CV
common test statistics	(F, t, χ^2, etc.)
confidence interval	CI
correlation coefficient (multiple)	R
correlation coefficient (simple)	r
covariance	cov
degree (angular)	°
degrees of freedom	df
expected value	*E*
greater than	>
greater than or equal to	≥
harvest per unit effort	HPUE
less than	<
less than or equal to	≤
logarithm (natural)	ln
logarithm (base 10)	log
logarithm (specify base)	log_2, etc.
minute (angular)	'
not significant	NS
null hypothesis	H_O
percent	%
probability	P
probability of a type I error (rejection of the null hypothesis when true)	α
probability of a type II error (acceptance of the null hypothesis when false)	β
second (angular)	″
standard deviation	SD
standard error	SE
variance	
population	Var
sample	var

SPECIAL PUBLICATION NO. 20-12

DOCUMENTATION OF CHINOOK, CHUM, AND COHO SALMON PRESENCE IN THE UPPER GOODPASTER RIVER GOLD MINING DISTRICT

by
Andrew D. Gryska
Alaska Department of Fish and Game, Sport Fish Division, Fairbanks

Alaska Department of Fish and Game
Division of Sport Fish, Research and Technical Services
333 Raspberry Road, Anchorage, Alaska, 99518-1565

December 2020

The Special Publication series was established by the Division of Sport Fish in 1991 for the publication of techniques and procedures manuals, informational pamphlets, special subject reports to decision-making bodies, symposia and workshop proceedings, application software documentation, in-house lectures, and similar documents. The series became a joint divisional series in 2004 with the Division of Commercial Fisheries. Special Publications are intended for fishery and other technical professionals. Special Publications are available through the Alaska State Library, Alaska Resources Library and Information Services (ARLIS), and on the Internet: http://www.adfg.alaska.gov/sf/publications/. This publication has undergone editorial and peer review.

Andrew D. Gryska,
Alaska Department of Fish and Game, Division of Sport Fish,
1300 College Road
Fairbanks, AK 99701, USA

This document should be cited as follows:
Gryska, A. D. 2020. Documentation of Chinook, chum, and coho salmon presence in the Upper Goodpaster River gold mining district. Alaska Department of Fish and Game, Special Publication No. 20-12, Anchorage.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF APPENDICES

ABSTRACT

Documentation and inventory of anadromous fish species has been limited in the Upper Goodpaster River drainage due to its remoteness. The mainstem Goodpaster River is known to have spawning populations of Chinook (*Oncorhynchus tshawytscha*) and chum (*O. keta*) salmon. Chinook salmon (*O. kisutch*) rear in the Goodpaster River but there is limited documentation regarding the extent of the drainage they utilize. Coho salmon spawn in the nearby Tanana and Delta Clearwater rivers, and it is possible juveniles rear in the Goodpaster River, although the latter has not been observed. This study proposed to document and list Pacific salmon *Oncorhynchus* sp. spawning and rearing habitat in the Upper Goodpaster River drainage using environmental DNA (eDNA) and capture of adults and juveniles. All findings were submitted to the Anadromous Waters Catalog.

Keywords: Tanana River, Goodpaster River, eDNA, Pacific salmon, *Oncorhynchus* spp.

INTRODUCTION

This study documented presence of Pacific salmon (*Oncorhynchus* spp.) habitat in previously undocumented areas of the Upper Goodpaster River (GPR) drainage. Documentation and inventory of anadromous fish species has been limited in the Upper GPR drainage due to difficulty in accessing some areas. Presently, there is a gold mine in the area (Pogo Mine), as well as numerous mineral claims that have potential for development. Identifying previously unknown areas of the drainage that provide habitat to anadromous fish will afford those areas greater protections and foster environmentally sensitive development of mineral extraction industries.

The GPR is a relatively large tributary (211 km long and 3,890 km^2 of drainage area) of the Tanana River, which is inaccessible by road except at one location at the Pogo mine (Figure 1). The river has anadromous populations of Chinook *O. tshawytscha* and chum salmon *O. keta*, and it may harbor juvenile coho salmon *O. kisutch,* although that has never been documented. It also has populations of resident fish such as Arctic grayling *Thymallus arcticus*, slimy sculpin *Cottus cognatus*, round whitefish *Prosopium cylindricum*, burbot *Lota lota*, and northern pike *Esox lucius*. Within the Upper GPR watershed, small-scale gold mining has occurred for over 100 years. During the 1990s, successful gold exploration occurred and ultimately resulted in construction of the Pogo hard rock mine facility, which is a large infrastructure mine with over 400 employees.

Prior to construction of the Pogo mine, Teck Cominco (the original owner) and the Alaska Department of Fish and Game (ADF&G) agreed to proactively fund and collect baseline data on Arctic grayling and Chinook salmon for use in future monitoring of population abundance and ecosystem health. Beginning in 2004, a Chinook salmon counting tower located just upstream of the South Fork GPR (Figure 1) has been operated by staff from Tanana Chiefs Conference (TCC) and the Bering Sea Fishermen's Association. Annual escapement counts have ranged from 723 to 4,280 between 2004 and 2015 (Table 1), but unlike the Chena and Salcha rivers, the GPR does not have an escapement goal. The estimates of escapement produced by the counting tower are considered representative of the GPR because the South Fork GPR has been deemed to have unsuitable spawning habitat for the majority of its drainage and several surveys have shown little, if any, spawning occurring in the South Fork. Nonetheless, it is unknown how significant the South Fork GPR is for either spawning or rearing of the GPR Chinook salmon stock.

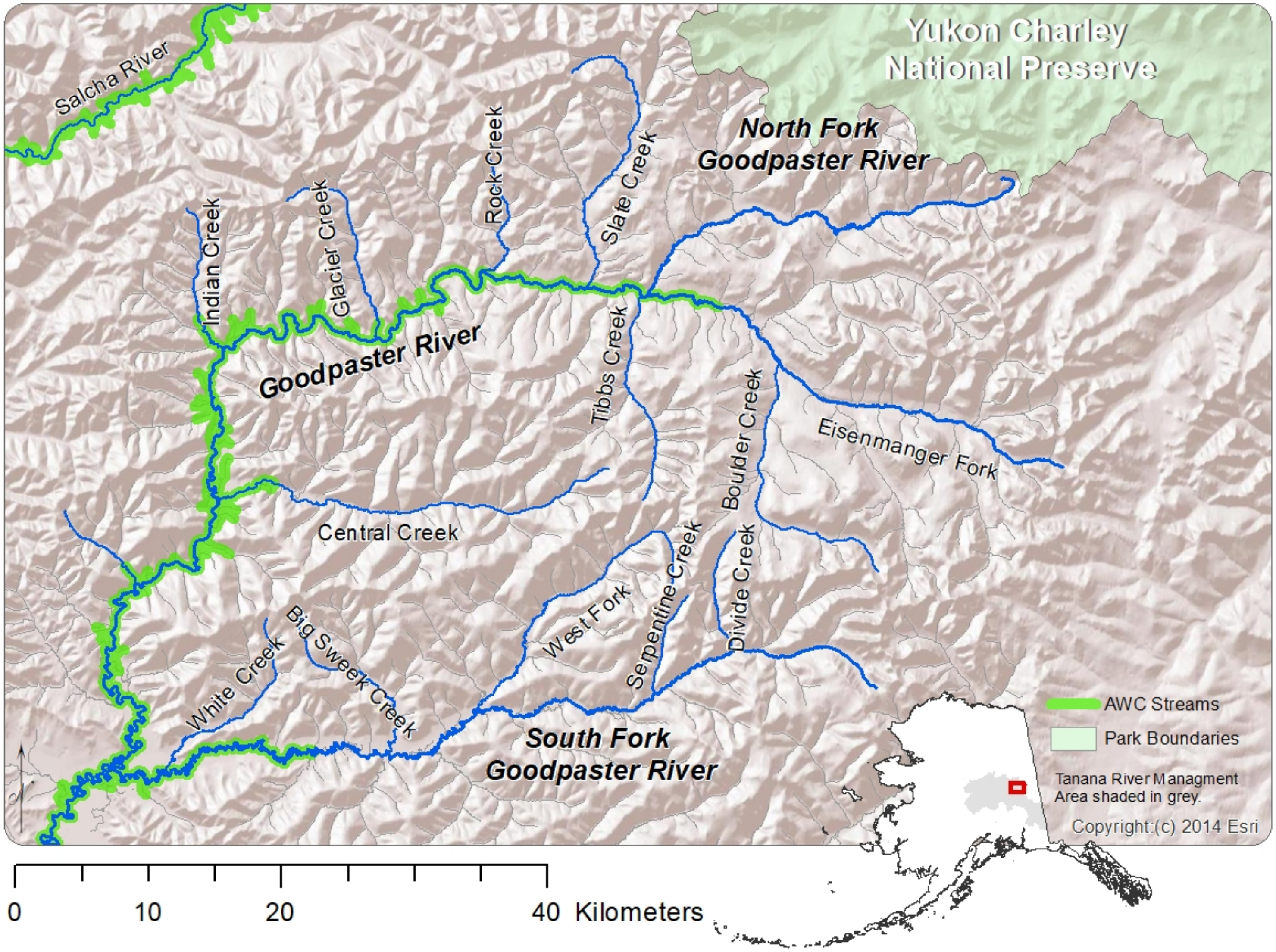

Figure 1.–Map of the Goodpaster River study area, 2017 regulatory AWC listed streams, and possible sample tributaries.

Table 1.–Estimates of the Goodpaster River Chinook salmon escapement (and SE), 2004–2015.

	Escapement	
Year	Estimate	SE
2004	3,673	106
2005	1,184	70
2006	2,479	100
2007	1,581	82
2008	1,880	85
2009	4,280	167
2010	1,167	67
2011	1,325	Not Reported
2012	752	50
2013	723	44
2014	1,305	90
2015	2,353	97

Currently, the mainstem GPR and small portions of some tributaries have documented presence of anadromous fishes (Chinook and chum salmon) listed in the Anadromous Waters Catalog (AWC) and its Atlas (Figure 1). ADF&G is required by Alaska Statute 16.05.871(a) to "specify the various rivers, lakes, and streams or parts of them" that are important to the spawning, rearing, or migration of anadromous fishes. By regulation 5AAC 95.011, the Anadromous Waters Catalog (AWC) and its associated Atlas list all documented salmon spawning and rearing areas. Documenting salmon spawning and rearing areas is important for the protection of their habitat and the conservation of the species. To add water bodies to the AWC and Atlas, a positive visual identification is required during biological fish surveys. These additions help managers evaluate sustainable harvest levels, protect critical habitats, and design future stock assessments.

Due to the overwhelming extent and remoteness of drainages in Alaska, it is difficult to sample each water body rigorously enough to be confident that salmon may or may not be present. Recently, a technique for detecting the presence of salmonids has been developed that utilizes environmental DNA (eDNA). Environmental DNA refers to the DNA molecules present in environmental samples such as water that were shed by organisms present in the environment. eDNA can originate from shedding of such things as skin cells and intestinal cells in feces or even from feces of birds and mammals that have consumed fish (Merkes et al. 2014). Assays of eDNA are a rapid, noninvasive method to quickly assess presence/absence of organisms in the local environment, such as a small tributary stream (Goldberg et al. 2011; Bohmann et al. 2014). The ability to rapidly assess the presence of salmon in any area of a drainage is a valuable tool; however, to be included in the AWC, a positive visual identification is required, and therefore eDNA evidence on its own is not suitable. Nonetheless, eDNA assay data can be used to efficiently focus future sampling efforts for the capture and documentation of salmon in areas where presence was detected. By quickly identifying salmon-bearing streams for inclusion in the AWC, not only are protections afforded to important fish habitat, but also permitting and development of projects such as mines can proceed more efficiently with appropriate planning.

OBJECTIVES

The research objectives for this study were to:

1. Identify important rearing and/or migration areas for juvenile and adult Chinook, chum, and coho salmon in remote tributaries of the Goodpaster River drainage for inclusion in the AWC.

METHODS

SAMPLE DESIGN

Documentation and verification of salmon presence in the Upper GPR was conducted over 2 open water seasons (2018 and 2019). During the first year, eDNA assays were used to document the presence or absence of salmon DNA in the environment, and during the second year, sites where eDNA indicated the presence of salmon were sampled to obtain live specimens. Sampling during 2018 consisted of collecting water samples for eDNA analysis (described below) at 31 locations within the drainage. DNA is extracted from filtered water samples, and the isolated DNA can then be used in species-specific assays where gene sequences are targeted for detection and used to estimate presence of a species. To examine the upstream extent of salmon presence, 2 or more water samples (near the mouth and at additional locations moving upstream) were collected on larger tributaries. Positive results from an eDNA sample could not identify life stage (juveniles or adults), and the results were not in and of themselves adequate justification for listing a species in the AWC. As such, verification for the AWC required a positive visual identification. For locations with positive eDNA results in 2018, each location was sampled during 2019 by using baited minnow traps and electrofishing to capture and identify salmon species and life stage. During 2019, additional sites were sampled for eDNA, and eDNA sample collections were repeated at some 2018 sample sites.

Project Area

The mainstem GPR is 211 km long and has a drainage area of 3,890 km^2 (Figure 1). The river is a rapid runoff stream that ranges from clear to slightly tannin stained, and it becomes turbid during periods of heavy runoff (Tack 1980). Upstream of the Pogo mine (river kilometer 109), the GPR has a drainage area of 1,753 km^2, and a mean summer stream flow of 17.0 m^3/s and a mean winter flow of 1.7 m^3/s (1997 to 2002). The mainstem GPR is accessible by road at only one location (Pogo mine), but it is consistently accessible by river boat up to Indian Creek (river kilometer 123). All other parts of the drainage are only accessible by helicopter or at several airplane landing strips.

Documentation of Presence

During 2018, 31 potential rearing sites in tributaries of the GPR were sampled for the presence of salmon eDNA for 3 days on August 29, August 30, and September 7. Samples for eDNA analysis were collected near the mouth of each tributary stream, and for larger tributaries additional samples were collected upstream of the mouth. During summer 2019, 18 sites were sampled for 4 days on August 12, 14, 18, and 19. Six sites that had positive eDNA for salmon in 2018 were sampled with minnow traps and electrofishing (and for eDNA at 2 sites), and 12 other sites were sampled for eDNA only.

Environmental DNA Procedures

Field methods for eDNA collection followed sample protocol of Carim et al. (2016), which has been demonstrated to be effective at eliminating contamination of samples (Appendix A). Water samples were collected using sterile equipment a minimum of 10 m upstream of a tributary confluence with a mainstem mixing area to avoid potential contamination from water in the main stream channel. Using a battery-powered peristaltic pump, 5 liters of water from the stream were pumped through a 47 mm, 0.45 µm cellulose nitrate filter (Whatman International Ltd., Little Chalfont, United Kingdom). Using sterile forceps, the filters were folded into quarters, rolled, and placed in a sample bag with silica desiccant beads. Sample bags were labeled with the same alphanumeric nomenclature as on data sheets.

At the end of the sampling period they were transported to a lab refrigerator until they were mailed to the National Geomic Center for Wildlife and Fish Conservation (NGC; U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station, Missoula, MT) for processing. Because eDNA samples can degrade over time, all samples were processed and mailed to NGC within 2 weeks of collection. Each individual sample was analysed for 3 species (Chinook, chum, and coho salmon). The analysis of eDNA was usually run in triplicate (3 wells). After analysis was complete, the results were returned with a description of presence or absence of DNA for each of the 3 salmon species in each sample and a quantitave measurement of the amount of DNA in the sample (#positive/#wells). This information was used to prioritize sampling areas during the second year of the study including select tributaries.

Capture Procedures

For sites where an eDNA sample indicated the presence of salmon, capture of salmon was attempted the following year. Chinook salmon juveniles were the primary species targeted as they overwinter in freshwater for 1 to 2 years before migrating out to sea, and they can be effectively captured in minnow traps. Juvenile chum salmon were not expected to be present in the fall given their life history of migrating to the ocean by early summer after emerging from the gravel as fry. To capture fish, 5 to 9 minnow traps were set for between 5 and 30 minutes at each sample site. Traps were distributed 100-300 m along the stream, and they were placed in spots that looked ideal for holding juvenile chinook salmon (deeper than 25 cm with cover such as woody debris and undercut banks). Captured juvenile fish were placed in an aerated tub, identifed to species, and measured for total length (TL), and a subsample of fish were photographed for documentation of species identification. After handling, fish were released alive.

Anadromous Waters Catalog nominations

For all sample areas where salmon were captured and identified by species, the stream was nominated for the AWC. Nominations for additions to the AWC are accepted year-round with an annual formal call from Sept. 1 through Sept. 30. Nominations may be submitted for the following purposes:

1. adding new streams,
2. adding species to cataloged streams,
3. extending species distribution in cataloged streams,
4. deleting streams or parts of them,
5. updating survey data on cataloged streams, or
6. revising stream channels, labeling errors, or identifying barriers to fish movement.

DATA COLLECTION

During the fieldwork, data was recorded in all-weather field notebooks and on each sample collected. GPS coordinates were recorded for all sites sampled. For each minnow trap set, recorded data was date, number of each species caught per trap, measurement of fish length to the nearest 1 mm FL, and location (river and GPS coordinates). Following all fieldwork, data items were transcribed into an EXCEL workbook for archival. Specifically, an EXCEL worksheet was made with column headings related to the field data form and comments. If anadromous species were captured, the species and location were submitted to the AWC using the required nomination form during the anuual call for nominations in September.

RESULTS

During 2018 sampling, 11 of 31 sampled sites strongly indicated Chinook salmon eDNA was present and 2 others also weakly indicated the presence of Chinook salmon (Table 2; Figure 2; Appendix B). During 2019, there was an attempt to capture Chinook salmon at each location that had indicated their presence in 2018. Due to an extremely rainy August, it was difficult to execute the project when and where planned, and unfortunately, high water prevented access to 7 of 13 sites (Table 3; Figure 3; Appendix B). Chinook salmon were captured at each site using minnow traps set for between 5 and 30m. eDNA samples were also collected at 14 sites of which 2 samples occurred at trapping sites (Table 3; Figure 3; Appendix B). Among the 2019 eDNA samples sites, 3 indicated presence of Chinook salmon, including the 2 trapping locations where Chinook salmon had been captured. Unlike the previous year, chum salmon eDNA was detected at 5 sites.

DISCUSSION

This study successfully identified 6 locations that harbored rearing Chinook salmon, which extended the known range of Chinook salmon in the GPR (Table 3; Figure 3). Each location was nominated to the AWC in September 2019 (Appendix C). An additional 7 locations had indications of Chinook salmon presence from eDNA results of 2018, but sampling crews were unable to access to these locations due to extreme rain and high-water conditions during August 2019. Additional eDNA sampling during 2019 was conducted at sample sites that were at or near 2018 samples sites, and in most cases, eDNA results reaffirmed the previous year's results. The exceptions were an indication of Chinook salmon presence in the West Fork of the South Fork GPR and an indication of chum salmon presence at 5 sites (Table 3).

The presence of chum salmon eDNA at 5 sites was beyond their known and expected distribution. For chum salmon, 4 of those 5 sites had low levels of eDNA (1 of 3 or 1 of 9 wells tested positive), which is indicative of low abundance. Juvenile chum salmon were not expected to be at the sample locations because they smolt to the ocean shortly after emerging from gravels in late spring and early summer (Salo 1991). Given high-water conditions, adult fish may have strayed into small tributary streams located upstream of typical spawning habitats. It is also conceivable other animals (such as a bird or bear) transported salmon carcasses, or deposited feces after consumption of salmon in the stream or within the stream drainage (Merkles et al. 2014). These sample sites were only 7 to 32 linear km from locations previously documented to have spawning chum salmon, close enough to regard the scenarios above as plausible.

The eDNA survey proved to be useful in directing successful efforts to capture Chinook salmon juveniles, but this study also revealed that improvements in sampling technique could be implemented to increase efficiency of limited resources (personnel and chartered helicopter). It was found that baited minnow traps captured Chinook salmon within 30 minutes, and that a total of 1 to 1.5 hours at a site was sufficient to sample with traps, electrofish, and collect the eDNA sample. Therefore, it is recommended the following protocol be used for future sampling of Chinook salmon. First, select sites upstream of known distribution and at multiple locations on larger tributaries. Second, set baited minnow traps for 30 minutes and electrofish a portion of stream (approximately 100–200 m). If fish collection procedures are unsuccessful, then collect eDNA at a point upstream of capture attempts so as to avoid contaminating sample collection. It may be necessary to collect eDNA at all sites if interested in the extent of chum and Coho salmon distribution beyond their known ranges. In the case of chum salmon, juveniles are unlikely to be captured in minnow traps. Coho parr may be captured in traps, but for the GPR, spawning areas are quite distant and any parr venturing so far up GPR tributaries are expected to be rare at best. When fish are rare, eDNA can be a useful tool for identifying where additional sampling may be required to document presence of salmon and can also yield indications of the extent of fish distributions.

Table 2.–Results of eDNA sampling in the Upper Goodpaster River drainage during 2018. Y indicates positive detection, (# positive wells/test wells) was an indication of how prevalent the eDNA was within the sample, and blank cells indicate no detection or capture.

Site #	Stream	Date	Chinook eDNA	Coho eDNA	Chum eDNA
18-1	Eisenmenger 1, tributary to GPR	8/29/18	Y (3/3)		
18-2	Eisenmenger 2, tributary to GPR	8/29/18			
18-3	Tributary 1, tributary to Eisenmenger Creek	8/29/18	Y (3/3)		
18-4	Tributary 2, tributary to Eisenmenger Creek	8/30/18			
18-5	Tributary 3, tributary to Eisenmenger Creek	8/29/18			
18-6	Upper Boulder Creek, tributary to Eisenmenger Creek	8/30/18			
18-7	Lower Boulder Creek, tributary to Eisenmenger Creek	9/7/18	Y (3/3)		
18-8	NF GPR 1, tributary to GPR	8/29/18	Y (3/3)		
18-9	NF GPR 2, tributary to GPR	8/29/18	Y (3/3)		
18-10	NF GPR 3, tributary to GPR	8/29/18			
18-11	NF GPR 4, tributary to GPR	8/29/18			
18-12	Upper Tibbs creek tributary to GPR	8/29/18	Y (3/3)		
18-13	Lower Tibbs Creek tributary to GPR	8/18/19	Y (3/3)		
18-14	Slate Creek tributary to GPR	9/7/18	Y (3/3)		
18-15	Rock Creek tributary to GPR	9/7/18	Y (1/9)		
18-16	Unknown tributary, tributary to GPR	9/7/18			
18-17	Glacier, tributary to GPR	9/7/18	Y (3/3)		
18-18	Central Creek 1, tributary to GPR	9/7/18	Y (3/3)		
18-19	Central Creek 2, tributary to GPR	9/7/18			
18-20	Sonora Creek, tributary to Central Creek	9/7/18			

-continued-

Table 2.–Page 2 of 2.

Site #	Stream	Date	Chinook eDNA	Coho eDNA	Chum eDNA
18-21	California Creek, tributary to Central Creek	9/7/18			
18-22	SF GPR 1, tributary to GPR	8/30/18	Y (3/3)		
18-23	SF GPR 2, tributary to GPR	8/30/18			
18-24	SF GPR 3, tributary to GPR	8/30/18			
18-25	SF GPR 4, tributary to GPR	8/30/18			
18-26	SF GPR 5, tributary to GPR	9/7/18			
18-27	Big Swede Creek, tributary to SF GPR	9/7/18	Y (3/3)		
18-28[a]	White Creek, tributary to SF GPR	9/7/18	n/a	n/a	n/a
18-29	WF 1, tributary to SF GPR	8/30/18			
18-30	WF 2, tributary to SF GPR	9/7/18			
18-31	WF 3, tributary to SF GPR	9/7/18			

Note: GPR = Goodpaster River, SF = South Fork, WF = West Fork

[a] The sample was too inhibited to yield a result.

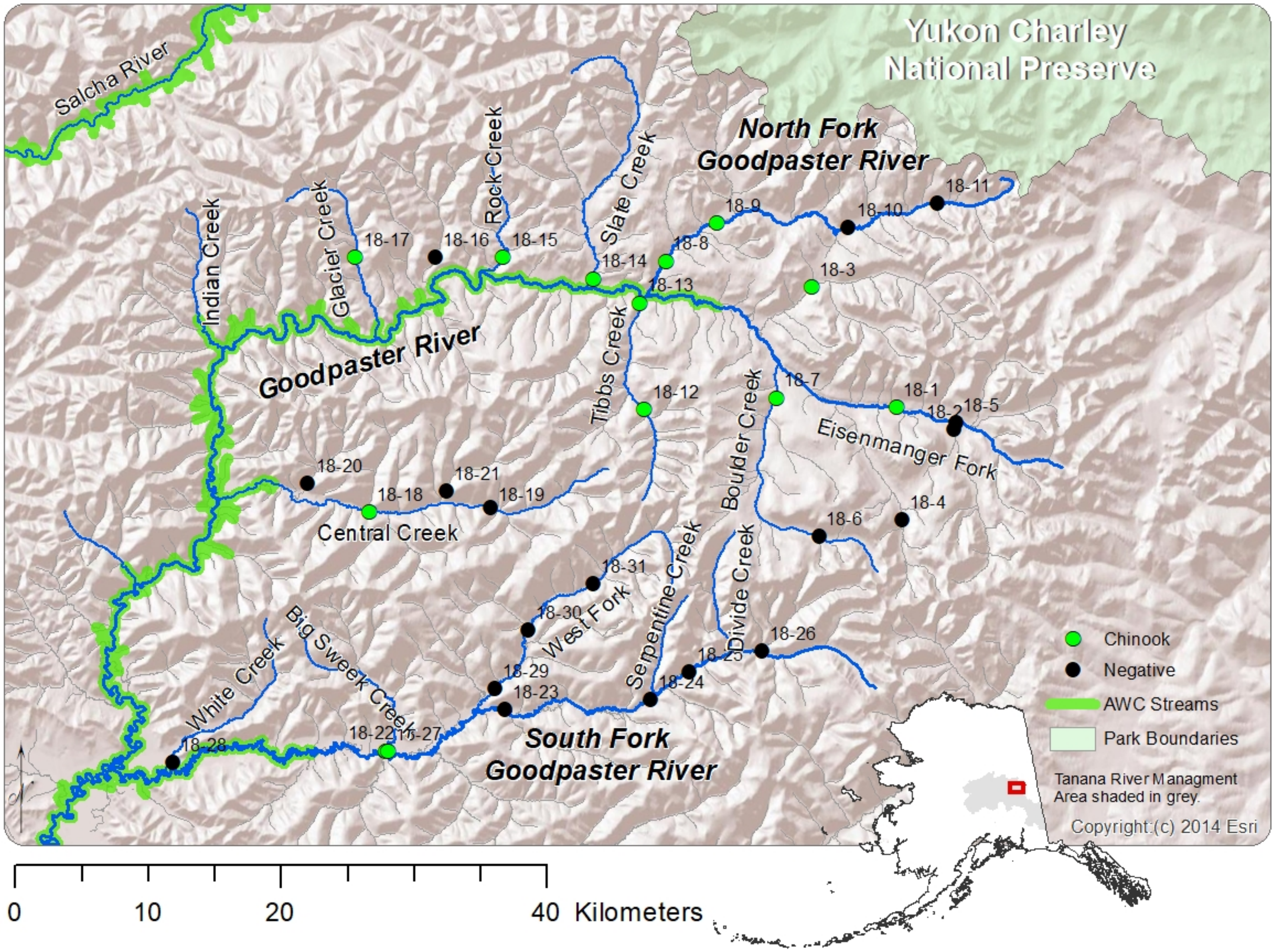

Figure 2.–Sample locations in the Upper Goodpaster River drainage during 2018 (labeled to correspond to Table 1).

Table 3.–Results of eDNA sampling and minnow trapping in the Upper Goodpaster River drainage during 2019. Y indicates positive detection or capture, (# positive wells/test wells) was an indication of how prevalent the eDNA was within the sample, and blank cells indicate no detection or capture.

Site #	Stream	Date	Chinook eDNA	Coho eDNA	Chum eDNA	Minnow Trap
19-1	Eisenmenger 2, tributary to GPR	8/19/19				
19-2	Tributary 1, tributary to Eisenmenger Creek	8/14/19	Y (3/3)			Y
19-3	Tributary 2, tributary to Eisenmenger Creek	8/19/19				
19-4	Tributary 3, tributary to Eisenmenger Creek	8/19/19			Y (3/3)	
19-5	NF GPR 2a, tributary to NF GPR	8/18/19			Y (1/3)	
19-6	NF GPR 5a, tributary to NF GPR	8/19/19				
19-7	North Fork Goodpaster 1, tributary to GPR	8/12/19	nc	nc	nc	Y
19-8	Lower Tibbs creek, tributary to GPR	8/12/19	nc	nc	nc	Y
19-9	Upper Tibbs Creek, tributary to GPR	8/18/19	nc	nc	nc	Y
19-10	Slate Creek, tributary to GPR	8/12/19	nc	nc	nc	Y
19-11	Glacier, tributary to GPR	8/14/19	Y (3/3)			Y
19-12	Central Creek 1a, tributary to Central Creek	8/18/09				
19-13	Sonora 2, tributary to Central Creek	8/19/19			Y (1/3)	
19-14	Central Creek 4, tributary to GPR	8/14/19			Y (1/9)	
19-15	SF GPR 6, tributary to GPR	8/14/19				
19-16	SF GPR 2, tributary to GPR	8/19/19				
19-17	WF GPR 1, tributary to SF GPR	8/19/19			Y (1/3)	
19-18	WF GPR 2a, tributary to WF of SF Goodpaster River	8/19/19	Y (1/3)			

Note: nc = not collected, GPR = Goodpaster River, NF = North Fork, SF = South Fork, WF = West Fork

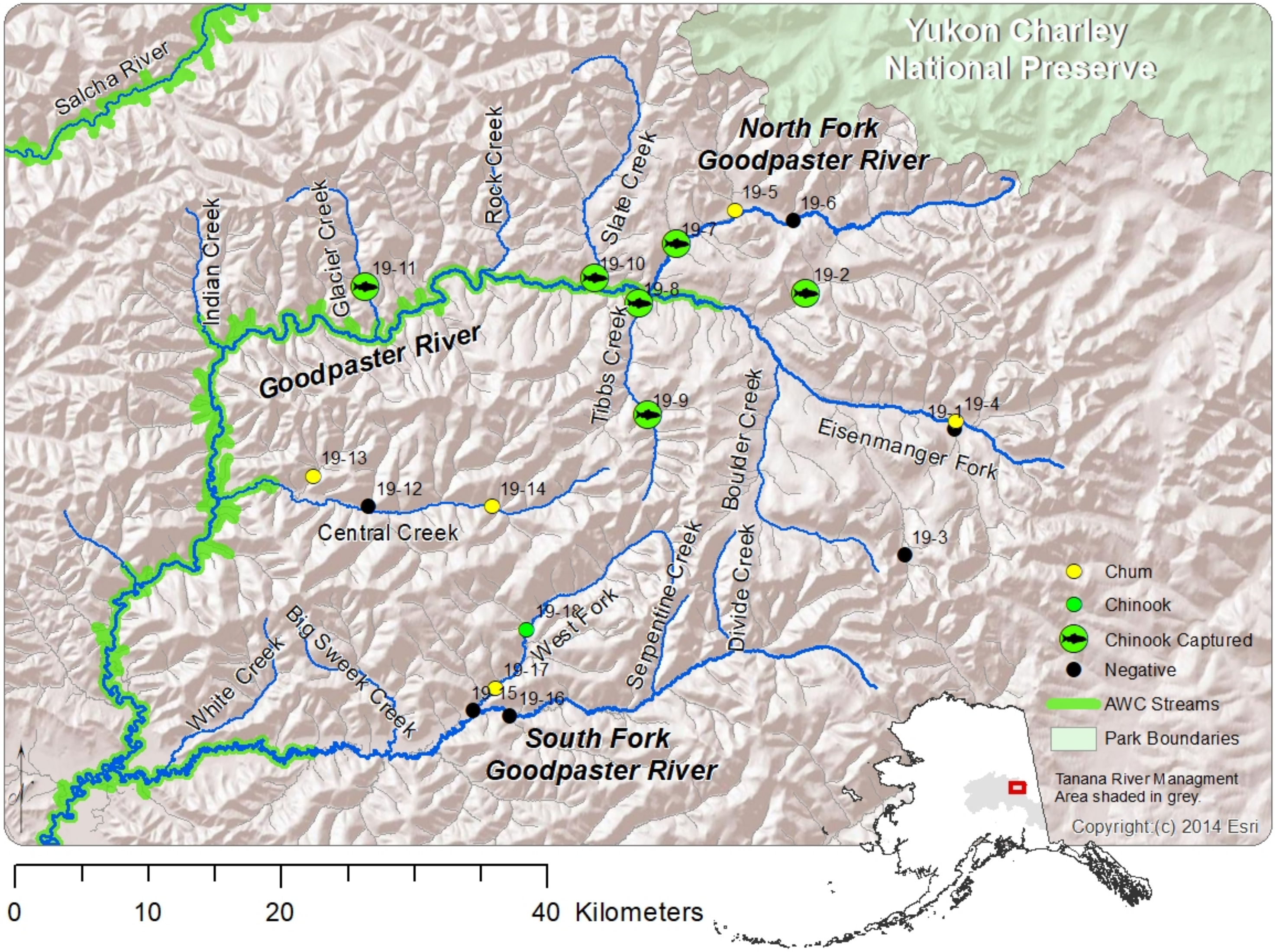

Figure 3.–Sample locations in the Upper Goodpaster River drainage during 2019 (labeled to correspond to Table 2).

ACKNOWLEDGEMENTS

The author thanks Sumitomo Metal Mining Pogo, LLC for providing funding and logistical support for this project. The sampling crew of Brian Collyard, April Behr, Melissa Osteen, Brandy Baker, and Eugene Petola are thanked for their assistance. Thanks to Tim Viavant, James Savereide, Phil Joy, and April Behr for their supervisory support, Rachael Kvapil for the editing and formatting of this report for publication, and Matt Tyers for his biometric review. An additional thanks to Keith Warren and pilots of Aurora Aviation Services for piloting sample surveys.

REFERENCES CITED

Bohmann, K., A. Evans, M. T. P. Gilbert, G. R. Carvalho, S. Creer, M. Knapp, D. W. Yu, and M. de Bruyn. 2014. Environmental DNA for wildlife biology and biodiversity monitoring. Trends in Ecology & Evolution 29:358-367.

Carim, K. J., K. S. McKelvey, M. K. Young, T. M. Wilcox, and M. K. Schwartz. 2016. A protocol for collecting environmental DNA samples from streams. General Technical Report RMRS-GTR-355. Fort Collins, CO: U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station.

Davis, A., and K. Ruddle. 2010. Constructing confidence: rational skepticism and systematic enquiry in local ecological knowledge research. Ecological Applications 20(3):880-894.

Goldberg, C. S., D. S. Pilliod, R. S. Arkle, and L. P. Waits. 2011. Molecular detection of vertebrates in stream water: a demonstration using Rocky Mountain tailed frogs and Idaho giant salamanders. PLoS ONE 6:e22746.

Merkes, C.M., S. G. McCalla, N. R. Jensen, M. P. Gaikowski, J. J. Amberg. 2014. Persistence of DNA in carcasses, slime and avian feces may affect interpretation of environmental DNA data. PLoS ONE 9(11):e0113346.

Salo, E. O. 1991. Life history of chum salmon (*Oncorhynchus keta*). Pages 231-310 [*In*] Groot, C., and L. Margolis, editors. Pacific Salmon life histories. University of British Columbia Press, Vancouver, BC, Canada.

Tack, S. L. 1980. Distribution, abundance, and natural history of the Arctic grayling in the Tanana River drainage. Alaska Department of Fish and Game, Federal Aid in Fish Restoration, Annual Performance Report, 1971-1980. Project F-9-12, 21(R-1).

APPENDICES

Appendix A.–eDNA Field Sampling Abbreviated Protocol as developed by the National Genomics Lab of the U.S. Forest Service.

National Genomics Center

eDNA Field Sampling Abbreviated Protocol

May 2018

This protocol is intended to be a field reference for the collection of eDNA samples. It does not cover the concepts of avoiding contamination and sampling site selection. Refer to the current General Technical Report (GTR) for the full protocol and troubleshooting guide.

Before Leaving for the Field

- Read full eDNA protocol (GTR: Carim et al. 2016) and the most recent field advisory.
- Charge pump batteries and check contents of duffel bag and pump case.
- Remember to clean pump, tubing, and bucket if moving between watersheds.

Reminders to Avoid Contamination

- Always keep personnel and gear downstream of sampling locations.
- Move upstream between sites while collecting samples.
- If you suspect that any gloves, forceps, filter cups, samples, etc. have become contaminated, immediately discard the item and start over with a new kit.

---

eDNA Sample Collection

1. Select an appropriate sampling site for your project objectives.
    - Sample downstream of suitable habitat. See GTR for details.
    - Beware of sampling in an eddy, backflow, splash pool, or anywhere in which flows might transport DNA from you onto the sample filter.
    - Make sure work area is free from hazardous obstacles.
2. Place pump and battery in stable area. Connect pump to battery using power cord.
3. Load tubing into the pump. Lift the lever on the pump head, load the tubing, then lower the lever. Follow the tubing orientation marked on pump and place outflow end of the tubing into the bucket.
4. Ensure pump speed is set to FAST and pump direction is set to FORWARD. Test suction from the filter adapter on your hand. If necessary to set down, place adaptor end of tubing down in an area where dirt/debris will not enter and clog the tubing.
5. Remove sampling kit from the "clean" white bag and immediately close the "clean" bag. Keep unused kits separate from working area to prevent contamination between sites.
6. Put on gloves from kit. Once wearing gloves do not touch anything that may be contaminated with DNA from outside the site (you, your pack, the pump, etc.).
7. Remove the packaged filter cup from the gallon kit bag. Without touching the filter cup, open the bag and rotate the filter cup within the Ziploc so that the blue base is facing towards the opening.
8. While holding the filter cup through the Ziploc, attach it to the tubing by pressing the blue base onto the tubing adapter. Do not fully remove the cup from packaging until ready to collect the sample. See photos on back.
    - If sampling by yourself, the hand that holds the tubing in now "dirty" and should not come into contact with the filter cup or "clean" sampling supplies for the remainder of sampling. See GTR for further details.

-continued-

9. Turn the pump on. While holding the filter cup by the adapter, lower the cup into the stream, keeping it in front of you and pointing upstream. In shallow streams, the tubing can be carefully lodged under a rock during filtering to hold it in place for the duration of filtering. Be sure not to pass your hand in front of the filter cup or dislodge rocks upstream of the filter cup.
    - To prevent contamination, keep all equipment and yourself downstream of the filter cup. Do not hold the cup by anything other than the adapter.
10. Filter 5 liters of water, measured by the 5 L mark in the outflow bucket. Filtering generally takes 8–10 minutes, but can take up to 20 minutes. If the amount of water being filtered slows considerably, the filter may be clogged.
    - If filter is clogged, follow protocol until step 15. Record the volume of filtered water on filter Ziploc. Continue from step 7 with a new filter until a total of 5 L of water has been filtered. However, do not exceed 3 filters per site, even if you do not reach 5 L.
11. Lift the cup upwards and away from the stream. Continue to run the pump for ~30 seconds to dry the filter.
12. Remove the cup from filter holder (squeeze and pull off, do not twist).
13. With the sterile forceps, fold the filter in quarters, with the filtering side in. **Do not touch the forceps to anything other than the filter.**
14. Place folded filter into the Ziploc containing silica beads. Do not allow anything to come in contact with the inside of the silica bag except for filter (watch contact with hands). Make sure filter is touching the silica beads, remove excess air, and seal completely.
15. For each filter, clearly label the Ziploc with the date, stream name, site, GPS coordinates, volume filtered, and your initials.
16. Turn off pump before placing tubing on the ground to avoid clogging it with dirt/debris.
17. Return filter cup and forceps back into the gallon Ziploc, and place in black "used equipment" bag. Discard gloves and other Ziplocs directly into black bag.
18. Fill out the envelope with the same sample information as recorded in step 15. In addition, please add the total number of filters used at the site and their respective volumes. Duplication of this data from the Ziploc is necessary to prevent data loss.
19. Seal the Ziploc(s) in the labeled envelope and store separately from sampling materials to prevent cross-contamination. Store out of direct sunlight, free from excessive weight or handling, and extreme heat (i.e. don't leave in your field truck).
20. Discard the water from the outflow bucket.

GTR Citation: Carim, Kellie J.; McKelvey, Kevin S.; Young, Michael K.; Wilcox, Taylor M.; Schwartz, Michael K. 2016. A protocol for collecting environmental DNA samples from streams. Gen. Tech. Rep. RMRS-GTR-355. Fort Collins, CO: U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station. 18 p.

-continued-

2018 Field Season Advisory

Before the field:

- All technicians should fully read the GTR and the new abbreviated protocol (included) before sampling. GTR available at https://www.fs.usda.gov/treesearch/pubs/52466 or on request.
- Site # should be less than 8 characters, preferably 2 digits per stream. See example below.
- Please note that "sample" refers to a unit of analysis. Filter refers to the components of that sample. There can be multiple filters used per sample in order to reach the targeted 5 L filter volume.
    - Writing "sample 2" on an envelope/Ziploc that has the same site #/ID as another sample indicates it is a replicate to be analyzed and charged for separately. Use "filter b" to denote a second component of a sample.

Sampling procedure reminders:

- It can take up to 20 minutes to filter 5 L. Be patient and only collect additional filters if absolutely necessary. Be sure the speed control is turned up (turn knob to the right). If using a geo pump II, make sure the pump head is loaded on the "Series 2 Drive".
- If punctures form in the filter, discard the punctured filter and begin sampling with a new filter.
- Fold filters into quarters to protect the DNA on the filter from abrasion or damage.
- Sterilize pumps, tubing, bucket and other personal field equipment with 10% bleach if crossing watersheds to prevent transfer of invasive species.

Sample Labeling and Storage:

- Upon collecting a sample, immediately label both the Ziploc **and** envelope while still at your site, and then place the Ziploc in the envelope.
    - The envelope protects DNA from UV degradation and reduces punctures.
    - Redundancy in labeling prevents data loss, particularly from smudging, mismatching data, and lack of identifying information.
    - Labels should include all of the following information: date, sample identification (e.g., stream name and sample number), GPS coordinates (Lat/Long or UTM, not Degrees, Minutes, Seconds please), number and volume of filters sampled, your initials, and any relevant notes.
    - If you have not been provided with premade envelopes, follow these examples:

Standard	Bull Trout Inventory
Site #02 Example Creek 46.8787, -113.9966 6/1/2016 Collector: TF Volumes: filter a: 2L, filter b: 3L Notes: Here are some notes.	123-1 Example Creek UTM: 12 271673, 5196044 6/1/2016 Collector: TF Volume: 5L Notes: Here are different notes.

- There is only enough silica in each Ziploc to absorb the water from **one** filter. Keeping filters separate maintains DNA integrity.
- Samples collected with multiple filters should be submitted together. Three Ziplocs fit in each envelope, use a second clearly labeled envelope if necessary.
- If using old sampling kits and the silica is not bright orange, it will need to be replaced.
- Keep samples out of the sun or heat (like the dashboard/seat of a truck) to prevent DNA degradation. Within backpacks or coolers is sufficient. If you choose to store your samples in a cooler with ice, please take precautions to ensure the samples stay dry. Water will degrade DNA on the sample and may compromise results.

Thank you! Following these helps us process your samples quickly and keeps costs low.

Appendix B.–Sample site number, name, location, and date for 2018 samples.

Site #	Stream	Date	Latitude	Longitude
18-1	Eisenmenger 1, tributary to GPR	8/29/18	64.3711	-143.8871
18-2	Eisenmenger 2, tributary to GPR	8/29/18	64.3504	-143.8037
18-3	Tributary 1, tributary to Eisenmenger Creek	8/29/18	64.4597	-143.9891
18-4	Tributary 2, tributary to Eisenmenger Creek	8/30/18	64.2955	-143.9046
18-5	Tributary 3, tributary to Eisenmenger Creek	8/29/18	64.3549	-143.7988
18-6	Upper Boulder Creek, tributary to Eisenmenger Creek	8/30/18	64.2927	-144.0363
18-7	Lower Boulder Creek, tributary to Eisenmenger Creek	9/7/18	64.3895	-144.0692
18-8	NF GPR 1, tributary to GPR	8/29/18	64.4917	-144.2094
18-9	NF GPR 2, tributary to GPR	8/29/18	64.5121	-144.1205
18-10	NF GPR 3, tributary to GPR	8/29/18	64.4956	-143.9187
18-11	NF GPR 4, tributary to GPR	8/29/18	64.5033	-143.7747
18-12	Upper Tibbs creek tributary to GPR	8/29/18	64.3949	-144.2763
18-13	Lower Tibbs Creek tributary to GPR	8/18/18	64.4662	-144.2601
18-14	Slate Creek tributary to GPR	9/7/18	64.4870	-144.3256
18-15	Rock Creek tributary to GPR	9/7/18	64.5103	-144.4603
18-16	Unknown tributary, tributary to GPR	9/7/18	64.5170	-144.5661
18-17	Glacier, tributary to GPR	9/7/18	64.5241	-144.6901
18-18	Central Creek 1, tributary to GPR	9/7/18	64.3528	-144.7236
18-19	Central Creek 2, tributary to GPR	9/7/18	64.3448	-144.5353
18-20	Sonora Creek, tributary to Central Creek	9/7/18	64.3777	-144.8125
18-21	California Creek, tributary to Central Creek	9/7/18	64.3598	-144.6007
18-22	SF GPR 1, tributary to GPR	8/30/18	64.1917	-144.7493
18-23	SF GPR 2, tributary to GPR	8/30/18	64.2084	-144.5581
18-24	SF GPR 3, tributary to GPR	8/30/18	64.2011	-144.3328

-continued-

Appendix B.–Page 2 of 2.

Site #	Stream	Date	Latitude	Longitude
18-25	SF GPR 4, tributary to GPR	8/30/18	64.2159	-144.2673
18-26	SF GPR 5, tributary to GPR	9/7/18	64.2225	-144.1508
18-27	Big Swede Creek, tributary to SF GPR	9/7/18	64.1915	-144.7471
18-28	White Creek, tributary to SF GPR	9/7/18	64.2037	-145.0798
18-29	WF 1, tributary to SF GPR	8/30/18	64.2233	-144.5690
18-30	WF 2, tributary to SF GPR	9/7/18	64.2590	-144.5048
18-31	WF 3, tributary to SF GPR	9/7/18	64.2837	-144.3950

Note: GPR = Goodpaster River, NF = North Fork, SF = South Fork, WF = West Fork

Appendix C.–Sample site number, name, location, and date for 2019 samples.

Site #	Stream	Date	Latitude	Longitude
19-1	Eisenmenger 2, tributary to GPR	8/19/19	64.3504	-143.8039
19-2	Tributary 1, tributary to Eisenmenger Creek	8/14/19	64.4565	-144.0010
19-3	Tributary 2, tributary to Eisenmenger Creek	8/19/19	64.2719	-143.9102
19-4	Tributary 3, tributary to Eisenmenger Creek	8/19/19	64.3553	-143.7994
19-5	NF GPR 2a, tributary to NF GPR	8/18/19	64.5188	-144.0918
19-6	NF GPR 5a, tributary to NF GPR	8/19/19	64.5064	-144.0034
19-7	North Fork Goodpaster 1, tributary to GPR	8/12/19	64.5023	-144.1890
19-8	Lower Tibbs Creek, tributary to GPR	8/12/19	64.4662	-144.2606
19-9	Upper Tibbs creek tributary to GPR	8/18/19	64.3911	-144.2728
19-10	Slate Creek, tributary to GPR	8/19/19	64.4877	-144.3245
19-11	Glacier Creek, tributary to GPR	8/14/19	64.5039	-144.6819
19-12	Central Creek 1a, tributary to Central Creek	8/18/19	64.3566	-144.7246
19-13	Sonora 2, tributary to Central Creek	8/19/19	64.3819	-144.8033
19-14	Central Creek 4, tributary to GPR	8/14/19	64.3449	-144.5340
19-15	SF GPR 6, tributary to GPR	8/14/19	64.2105	-144.6068
19-16	SF GPR 2, tributary to GPR	8/19/19	64.2039	-144.5520
19-17	WF GPR 1, tributary to SF GPR	8/19/19	64.2234	-144.5688
19-18	WF GPR 2a, tributary to WF of SF Goodpaster River	8/19/19	64.2597	-144.5075

Note: GPR = Goodpaster River, NF = North Fork, SF = South Fork, WF = West Fork

Appendix D.–Anadromous Water Catalog Nomination form for Glacier Creek.

State of Alaska
Department of Fish and Game
Division of Sport Fish

Nomination Form
Anadromous Waters Catalog

Region: 2 Interior

USGS Quad(s): Big Delta

AWC Number of Water Body:

Name of Water body: Glacier Creek ☒ USGS Name ☐ Local Name

☒ Addition ☐ Deletion ☐ Correction ☐ Backup Information

For Office Use

Nomination # ______	______ Fisheries Scientist	______ Date
Revision Year: ______	______ Habitat Operations Manager	______ Date
Revision to: Atlas ____ Catalog ____ Both ____	______ AWC Project Biologist	______ Date
Revision Code: ______	______ GIS Analyst	______ Date

OBSERVATION INFORMATION

Species	Date(s) Observed	Spawning	Rearing	Present	Anadromous
Chinook salmon	8/14/2019		a	a	☒
					☐
					☐
					☐
					☐

IMPORTANT: Provide all supporting documentation that this water body is important for the spawning, rearing or migration of anadromous fish, including: number of fish and life stages observed; sampling methods, sampling duration and area sampled; copies of field notes; etc. Attach a copy of a map showing location of mouth and observed upper extent of each species, as well as other information such as: specific stream reaches observed as spawning or rearing habitat; locations, types, and heights of any barriers; etc.

Comments Glacier Creek, tributary of Goodpaster River, was sampled for eDNA on 9/7/2018 @ 64.52413 -144.69006. The eDNA sample was positive for Chinook salmon. The location was resampled 8/14/2019 @ 64.50391 -144.68183 using 6 minnow traps, soak time 10m. The traps captured 4 chinook salmon FL 76, 76, 79, 73 mm

Name of Observer (please print): Brian Collyard

Signature: Date: 9-30-19

Agency: ADF+G

Address: 1300 College Road
Fairbanks AK 99701

This certifies that in my best professional judgment and belief the above information is evidence that this waterbody should be included in or deleted from the Anadromous Waters Catalog.

Signature of Area Biologist: Heather Scanell Date: 9-30-19 Revision 11/13

Name of Area Biologist (please print): Heather Scanell

State of Alaska
Department of Fish and Game
Division of Sport Fish

Nomination Form
Anadromous Waters Catalog

Region ______ USGS Quad(s) ______

AWC Number of Water Body ______

Name of Water body ______ ☐ USGS Name ☐ Local Name

☐ Addition ☐ Deletion ☐ Correction ☐ Backup Information

For Office Use

Nomination # ______	______ Fisheries Scientist ______ Date
Revision Year: ______	______ Habitat Operations Manager ______ Date
Revision to: Atlas ____ Catalog ____ Both ____	______ AWC Project Biologist ______ Date
Revision Code: ______	______ GIS Analyst ______ Date

OBSERVATION INFORMATION

Species	Date(s) Observed	Spawning	Rearing	Present	Anadromous
					☐
					☐
					☐
					☐
					☐

IMPORTANT: Provide all supporting documentation that this water body is important for the spawning, rearing or migration of anadromous fish, including: number of fish and life stages observed; sampling methods, sampling duration and area sampled; copies of field notes; etc. Attach a copy of a map showing location of mouth and observed upper extent of each species, as well as other information such as: specific stream reaches observed as spawning or rearing habitat; locations, types, and heights of any barriers; etc.

Comments

Name of Observer (please print): ______

Signature: ______ Date: ______

Agency: ______

Address: ______

______

This certifies that in my best professional judgment and belief the above information is evidence that this waterbody should be included in or deleted from the Anadromous Waters Catalog.

Signature of Area Biologist: ______ **Date:**______ Revision 11/13

Name of Area Biologist (please print):

Appendix E.–Data files for eDNA collection and salmon capture in the in the Goodpaster River.

File name: Goodpaster River eDNA files for archive.xls

Note: Data files are archived at and are available from the Alaska Department of Fish and Game, Sport Fish Division, 1300 College Road, Fairbanks, Alaska 99701-1599.